

Name: Krishna RenjithDate: 5/29/2025Class: 4

MyDesign Portfolio

Element L Initial Template

Element L: Presentation of Designer's Recommendations

Content of Recommendations (L1, L2)

"Taking into account <something we did>, we believe doing _____ would improve _____ because _____."

- What could be done next to go further with this project?

Our project was set on fire and destroyed during the Burn Day Challenge. However, one thing which could be done to go further with this project includes rebuilding the house and detection system, and then making changes to our suppression system (sprinklers), which had failed during testing. Following the comments of our field expert, Dr. Jim Milke, we should have designed more sprinklers and filled them with more water and secured them to the ceiling to ensure that they would effectively put out the fire. Taking into account that our water balloons (sprinklers) were too small and did not hang from the ceiling, we believe filling the balloons with more water would improve the suppression system because there would be more than enough water needed to put out the fire.

- What would you keep if you did the project again? *Generate a consistently detailed list of what would be kept in the prototype or testing plans.*

If I did the project again, I would keep the Arduino Fire-Alarm detection system that one of our team members had built. This system worked extremely well, in that it detected the fire and began beeping loudly before most of the furniture or house was lit up. Taking into account that this detection system was tested several times using a lighter, we believe rebuilding it in the project would allow us to effectively detect a fire.

- What would you change in your current prototype or testing plan? *Generate a consistently detailed list of HOW these details would be implemented.*

One thing I would change in our current design would be to replace the small water

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balloons which acted as our sprinkler system with larger plastic bags filled with lots of water. After seeing how one group had failed to activate their suppression system (water balloons were too thick to melt open), our group realized that our balloons were too thin and too small. Next time, we should also ensure that our water balloons are stuck to the ceiling, since they fell off during the ride to the testing location and did not work at all.

- What would have made this project easier for you to complete? **Generate a consistently detailed list of things that would make the project easier.**

One thing which would have made this project easier for me to complete would have been working in a smaller group. My group had over five members, and since every member was not always present, certain parts of the project would be completed before others. As a result, it became difficult to finish the project on time, which caused a lot of stress as well as an unfinished suppression system. While working in a larger group had its benefits in that one member was good at coding the detection system while another could cut cardboard well, the chaos which ensued with ensuring each person completed their task and was present made the experience unpleasant and difficult to fulfill.

Step L: Presentation of Designer's Recommendations						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
L1. Content of recommendations. Recommendations include _____.	specific and detailed ways the project can be improved and a rationale for why they are needed	specific and detailed ways the project can be improved	some specific ways the project can be improved	one way the project can be improved	vague recommendation(s) for improvement of the project	no recommendations for improvement of the project
L2. Detailed plans for implementation of	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0

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improvements. Plans for implementation of improvements are _____ 	consistently detailed	generally detailed	minimally detailed and may include one or more caveats	partially detailed and/or overly general	vague/unclear or minimally related to the recommendations provided	not offered
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